

2011
**HIGHER SCHOOL CERTIFICATE
EXAMINATION**

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 13, 15, 19, 21 and 23

Total marks – 100

Section I Pages 2–24

75 marks

This section has two parts, Part A and Part B

Part A – 20 marks

- Attempt Questions 1–20
- Allow about 35 minutes for this part

Part B – 55 marks

- Attempt Questions 21–31
- Allow about 1 hour and 40 minutes for this part

Section II Pages 25–36

25 marks

- Attempt ONE question from Questions 32–36
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 20 marks

Attempt Questions 1–20

Allow about 35 minutes for this part

Use the multiple-choice answer sheet for Questions 1–20.

- 1 The DNA in plants and animals is composed of the same chemical components.
- What is the biological significance of this statement?
- (A) Plants and animals can interbreed.
 - (B) Plants and animals share common ancestry.
 - (C) Plants and animals are genetically identical.
 - (D) Plants and animals are composed of identical proteins.
- 2 Which substance is mainly bound to proteins when it is carried in mammalian blood?
- (A) Nitrogenous waste
 - (B) Carbon dioxide
 - (C) Lipid
 - (D) Salt
- 3 In organisms, the maintenance of a constant internal environment is
- (A) necessary because organisms must have a constant body temperature.
 - (B) necessary because enzyme activity is highest at specific temperatures.
 - (C) unnecessary because organisms are found in environments with a broad range of temperatures.
 - (D) unnecessary because the nervous system detects and responds to changes in ambient temperature.
- 4 Which of the following alternatives lists ALL of the chemical components of a chromosome?
- (A) Sugar, phosphate and bases
 - (B) Lipids, DNA and protein
 - (C) DNA and protein
 - (D) Genes and DNA

- 5 The presence of freckles is a dominant characteristic. A child's mother has no freckles and its father is heterozygous for freckles.

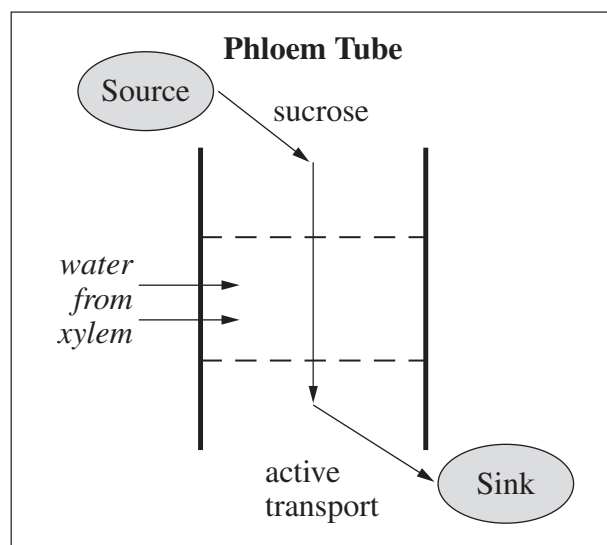
What is the probability that this child will have freckles?

- (A) 25%
 - (B) 50%
 - (C) 75%
 - (D) 100%
- 6 What was Macfarlane Burnet's contribution to our understanding of human disease?
- (A) He determined that mosquitoes transmit malaria.
 - (B) He disproved the theory of spontaneous generation.
 - (C) He developed a method for establishing that a particular microbe causes a disease.
 - (D) He discovered how the body distinguishes between its own tissues and foreign cells.
- 7 How do cleanliness in food preparation and cleanliness in personal hygiene assist in the control of disease?
- (A) They reduce the microflora.
 - (B) They reduce transmission of pathogens.
 - (C) They stop the transfer of dirt between people.
 - (D) They stop food that contains bacteria from being eaten.
- 8 Which type of cells destroy virally infected cells?
- (A) B cells
 - (B) Killer T cells
 - (C) Helper T cells
 - (D) Cytotoxic T cells
- 9 Which list shows pathogens in order of increasing size?
- (A) Bacterium, prion, virus, protozoan, macroparasite
 - (B) Macroparasite, prion, protozoan, bacterium, virus
 - (C) Prion, virus, bacterium, protozoan, macroparasite
 - (D) Virus, prion, bacterium, macroparasite, protozoan

- 10 Which of the following correctly identifies a source of variation in both asexual and sexual reproduction?

	<i>Asexual</i>	<i>Sexual</i>
(A)	Cloning	Natural selection
(B)	Crossing over	Cell division
(C)	Spontaneous generation	Fertilisation
(D)	Mutation	Mutation

- 11 The diagram represents one current theory for the movement of materials in phloem.



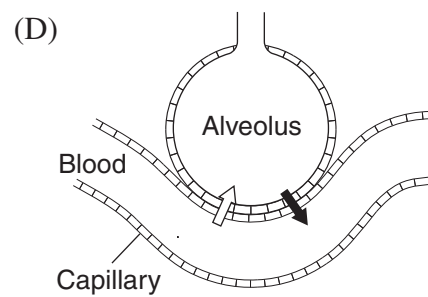
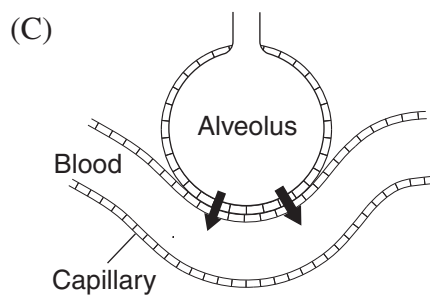
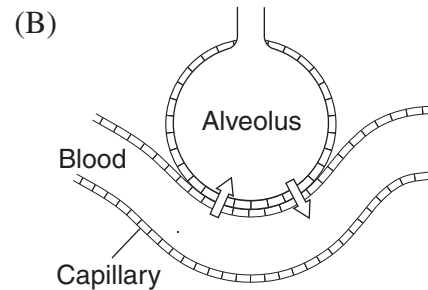
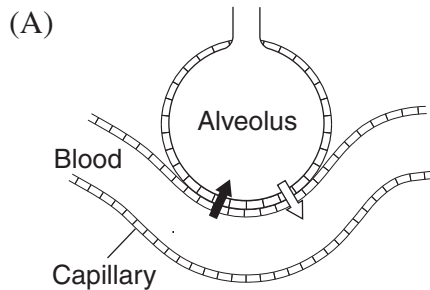
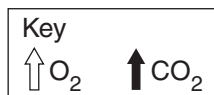
In the process illustrated in the diagram, water from xylem

- (A) causes a build up of pressure.
 - (B) allows adhesion for capillarity.
 - (C) provides water for photosynthesis.
 - (D) dilutes sucrose for active transport.
- 12 An Australian insect produces uric acid and no other form of nitrogenous waste.

What is the purpose of this adaptation?

- (A) To increase salt loss
- (B) To reduce water loss
- (C) To increase its toxicity to predators
- (D) To reduce the chance of attracting predators

13 Which diagram best explains changes to the composition of blood in the lungs?



14 What role do mucous membranes and cilia play in the body's defence against pathogens?

- (A) They cause cell death to seal off pathogens.
- (B) They form part of a barrier preventing entry of pathogens.
- (C) They remove pathogens as part of the inflammatory response.
- (D) They recognise pathogens passing through the lymphatic system.

15 African ostriches, South American rheas and Australian emus are all large extant flightless birds. These observations provide evidence for the theory of evolution.

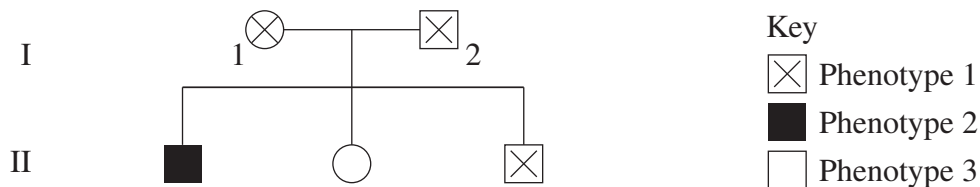
This evidence for evolution belongs to

- (A) biochemistry.
- (B) biogeography.
- (C) paleontology.
- (D) comparative embryology.

- 16 Which of the following shows the correct sequence of steps in the interaction between T cells (lymphocytes) and B cells (lymphocytes)?

	Step 1	Step 2	Step 3
(A)	T cells interact with antigens presented by macrophages	Activated T cells differentiate into helper T cells	Helper T cells produce chemicals that cause B cells to multiply
(B)	T cells interact with antigens presented by macrophages	Activated T cells differentiate into B cells	B cells produce chemicals that cause T suppressor cells to multiply
(C)	B cells interact with antigens presented by macrophages	Activated B cells differentiate into plasma cells	Plasma cells produce chemicals that cause T cells to multiply
(D)	B cells interact with antigens presented by macrophages	Activated B cells differentiate into B memory cells	B memory cells produce chemicals that cause macrophages to multiply

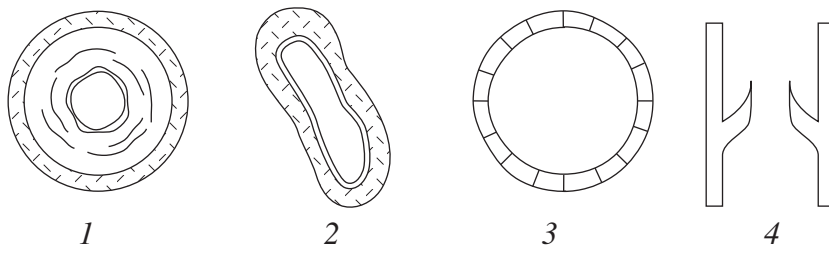
- 17 The pedigree shows the inheritance of a trait controlled by a pair of alleles.



Which Punnett square correctly represents the cross between the parents in generation I?

- (A)
- | | | |
|---|----|----|
| | B | b |
| A | AB | Ab |
| a | aB | ab |
- (B)
- | | | |
|---|----|----|
| | B | B |
| A | AB | AB |
| A | AB | AB |
- (C)
- | | | |
|---|----|----|
| | B | b |
| B | BB | Bb |
| b | Bb | bb |
- (D)
- | | | |
|---|----|----|
| | A | B |
| A | AA | AB |
| B | AB | BB |

18 Diagrams of blood vessels are shown (not to scale).



Which of the following correctly lists the names of the blood vessels shown?

	<i>1</i>	<i>2</i>	<i>3</i>	<i>4</i>
(A)	Artery	Vein	Capillary	Vein
(B)	Capillary	Artery	Artery	Vein
(C)	Artery	Vein	Vein	Capillary
(D)	Vein	Artery	Capillary	Capillary

19 Why are antibiotics ineffective in treating malaria?

- (A) Malaria is not caused by bacteria.
- (B) Malaria is not an infectious disease.
- (C) Malaria is transmitted by mosquitoes.
- (D) Malaria is associated with wet environments.

20 How does the fossil record provide evidence to support the concept of punctuated equilibrium?

- (A) The fossil record is incomplete.
- (B) The fossil record shows that some organisms have become extinct.
- (C) The fossil record shows that there are short periods of rapid change in fossil forms.
- (D) The fossil record shows that some organisms change gradually over geologic time.

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Biology

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Centre Number

Section I (continued)

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Student Number

Part B – 55 marks

Attempt Questions 21–31

Allow about 1 hour and 40 minutes for this part

Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.

Question 21 (4 marks)

- (a) Identify TWO methods used to treat drinking water to reduce the risk of infection. **2**

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- (b) Outline how the use of each of these methods reduces the risk of infection. **2**

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Question 22 (4 marks)

Six students performed a trial experiment on enzyme activity. The enzyme they were studying acts on a cloudy suspension, breaking it down into a soluble form. The lesson ended and students were asked to stop their experiment. They then recorded the time the experiment had run and their observations. These data are collated in the table.

<i>Student</i>	<i>Volume of enzyme solution (mL)</i>	<i>Temperature (°C)</i>	<i>pH</i>	<i>Volume of cloudy suspension (mL)</i>	<i>Time (minutes)</i>	<i>Observation at end of lesson</i>
1	5	37	2	10	9	Cloudy
2	5	37	2	10	20	Clear
3	5	37	4	10	35	Clear
4	5	37	4	10	20	Cloudy
5	5	37	6	10	8	Cloudy
6	5	37	6	10	40	Cloudy

- (a) What is the purpose of this experiment?

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- (b) Describe TWO changes that would improve the validity of the data collected in the experiment.

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Question 22 continues on page 11

Question 22 (continued)

- (c) The experiment was repeated with all trials running for 40 minutes. Complete the table to predict the results.

1

<i>Student</i>	<i>Volume of enzyme solution (mL)</i>	<i>Temperature (°C)</i>	<i>pH</i>	<i>Volume of cloudy suspension (mL)</i>	<i>Time (minutes)</i>	<i>Observation at end of trial</i>
1	5	37	2	10	40	
2	5	37	2	10	40	
3	5	37	4	10	40	
4	5	37	4	10	40	
5	5	37	6	10	40	
6	5	37	6	10	40	

End of Question 22

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Centre Number

Section I – Part B (continued)

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Student Number

Question 23 (5 marks)

Several scientists were involved in determining the structure of DNA.

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To what extent did the quality of collaboration and communication between these scientists impact on their scientific research?

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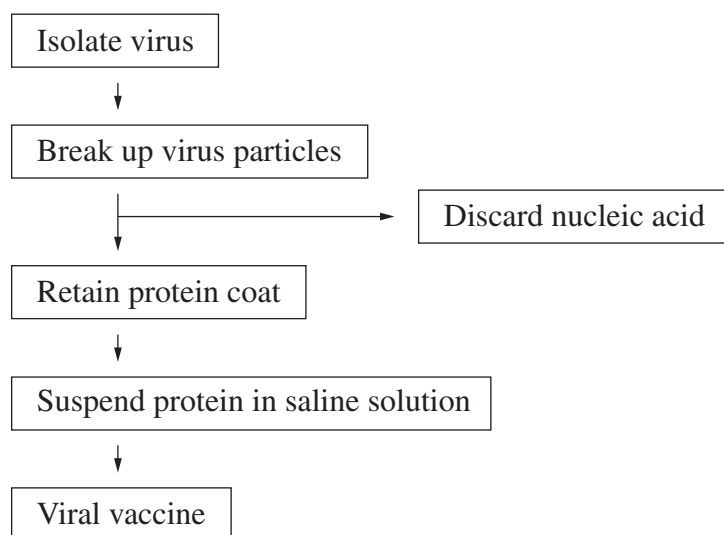
Question 24 (5 marks)

- (a) Name a viral disease that is controlled by the use of a vaccine.

1

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- (b) The flowchart shows the steps involved in the preparation of a vaccine to prevent a viral disease.



- (i) Outline how a vaccine that is prepared in this way will produce immunity to a specific viral disease.

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- (ii) Explain ONE procedure in the flowchart that ensures the safety of the prepared vaccine.

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2011 HIGHER SCHOOL CERTIFICATE EXAMINATION

Biology

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Section I – Part B (continued)

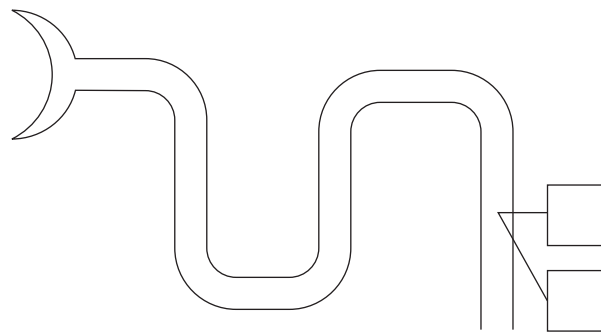
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Student Number

Question 25 (5 marks)

A diagram of a nephron is shown.

Processes that occur in a nephron

**Key:**

- A Filtration takes place
- B Water reabsorption takes place
- C Glucose reabsorption takes place
- D Antidiuretic hormone (ADH) has its effect

(a) Label each of the two boxes on the diagram using A, B, C or D to identify the processes that take place at this location. 2

(b) Which one of the above processes (A, B, C or D) occurs due to active transport? 1

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(c) Outline the effect of aldosterone on the control of body fluids. 2

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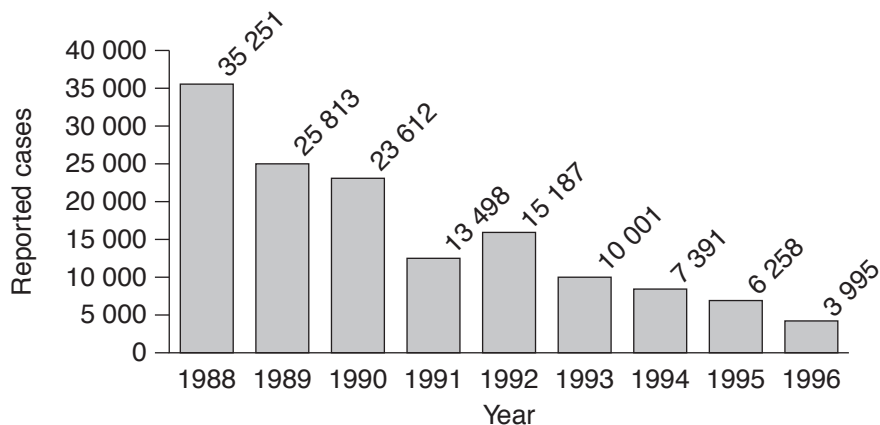
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Question 26 (6 marks)

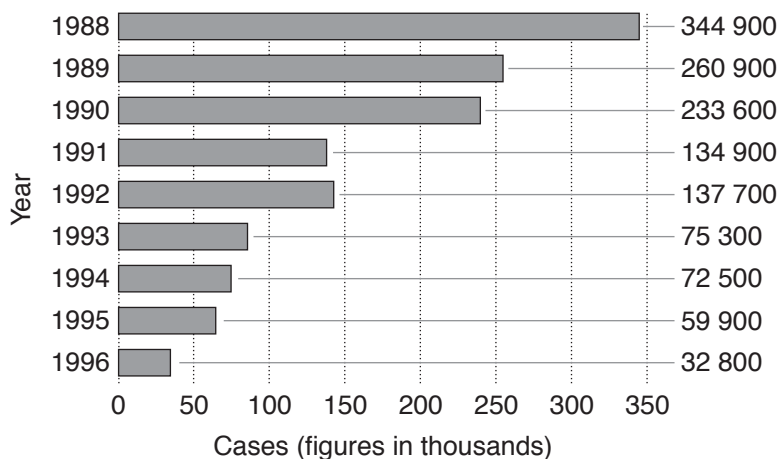
In assessing the effectiveness of the global polio vaccination program started in 1988, a student collected Graphs A and B from two reputable sources.

Graph A: Reported annual global polio cases from 1988–1996



Source: World Health Organization
<http://www.who.int/en/>
Date accessed: 10/6/201

Graph B: Estimated annual global polio cases from 1988–1996



Source: World Health Organization
<http://www.who.int/en/>
Date accessed: 10/6/2011

Question 26 continues on page 17

**Question 26 (continued)**

- (a) Identify TWO similarities and TWO differences between the information contained in Graphs A and B. **4**

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- (b) Recommend TWO additional types of data that could be used for evaluating the effectiveness of the global polio vaccination program. **2**

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End of Question 26



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Centre Number

Section I – Part B (continued)

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Student Number

Question 27 (4 marks)

A new non-infectious disease has been discovered in people in some aged care homes.

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Outline at least FOUR features of an epidemiological study that could be used to identify the cause of this disease.

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Question 28 (5 marks)

Stem cells can develop into any type of body cell. When stimulated they can differentiate into specialised cells.

The skin contains stem cells as well as epidermal, blood vessel, fat and muscle cells.

New research suggests that stem cells obtained from the skin of a patient can be used to make red and white blood cells for that patient in the laboratory. Production of new improved artificial blood seems a likely outcome.

- (a) Outline how gene expression in skin stem cells is linked to the maintenance of health after the skin is broken. **2**

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- (b) Explain features of the new artificial blood made from skin stem cells that are likely to be an improvement over an existing form of artificial blood. **3**

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Biology

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Centre Number

Section I – Part B (continued)

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Student Number

Question 29 (5 marks)

You are required to plan and perform a first-hand investigation to identify microbes in food or in water.

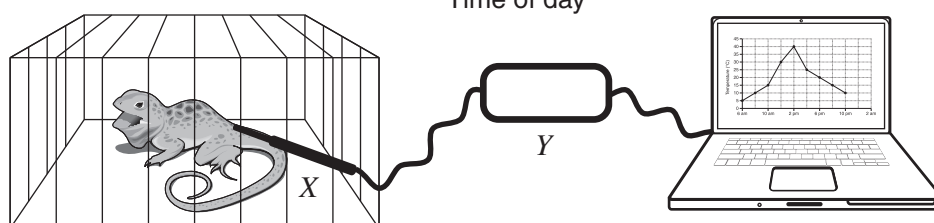
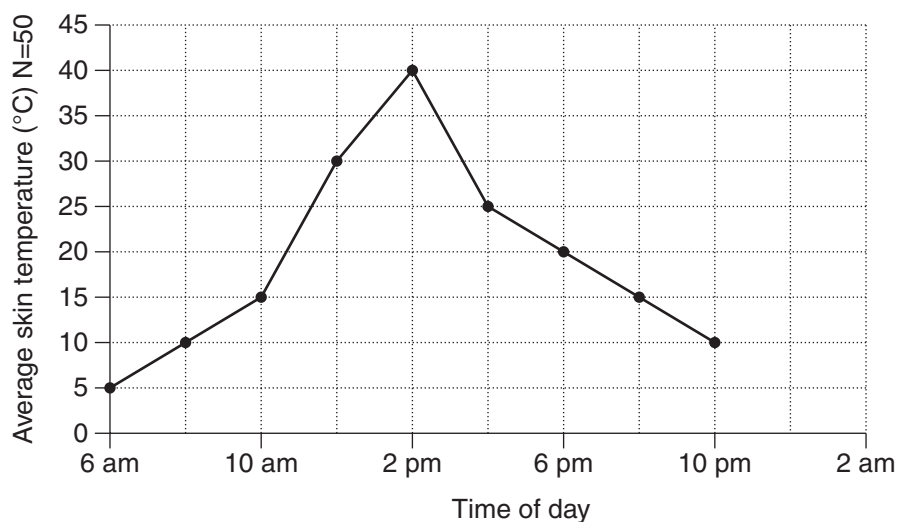
5

Complete the following table for your investigation.

Dependent variable	
Independent variable	
Control	
Safe work practices to be followed	

Question 30 (5 marks)

Scientists recorded the body temperature of 50 reptiles of the same species on the same day. They were kept in small cages in the shade in a hot desert habitat. Technology X was used to measure skin temperature. This was linked to technology Y and then to a computer as shown in the diagram. The graph shows the averaged data generated by the computer from this experiment.



- (a) Identify technology X and technology Y.

2

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- (b) The scientists concluded that the body temperature of this species of reptile is only controlled by the ambient temperature.

3

Construct an argument against this conclusion, based on the information provided.

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2011 HIGHER SCHOOL CERTIFICATE EXAMINATION

Biology

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Centre Number

Section I – Part B (continued)

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Student Number

Question 31 (7 marks)

Fungi are a natural source of antibiotics. A scientist developed a new antibiotic by exposing a fungus to radiation.

7

Information relevant to this antibiotic:

- It stops the activity of an enzyme in pathogenic bacteria.
- It has no effect on a similar enzyme in humans.
- The chemical composition of the enzyme in humans differs from the enzyme in the pathogenic bacteria by two amino acids.
- It is used to treat humans infected with the pathogenic bacteria.

Using this example and other relevant knowledge, describe how advances in our understanding of biology have implications for society.

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Question 31 continues on page 24

Question 31 (continued)

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End of Question 31

Biology

Section II

25 marks

Attempt ONE question from Questions 32–36

Allow about 45 minutes for this section

Answer parts (a)–(c) of the question in Section II Answer Booklet 1.

Answer parts (d)–(e) of the question in Section II Answer Booklet 2.

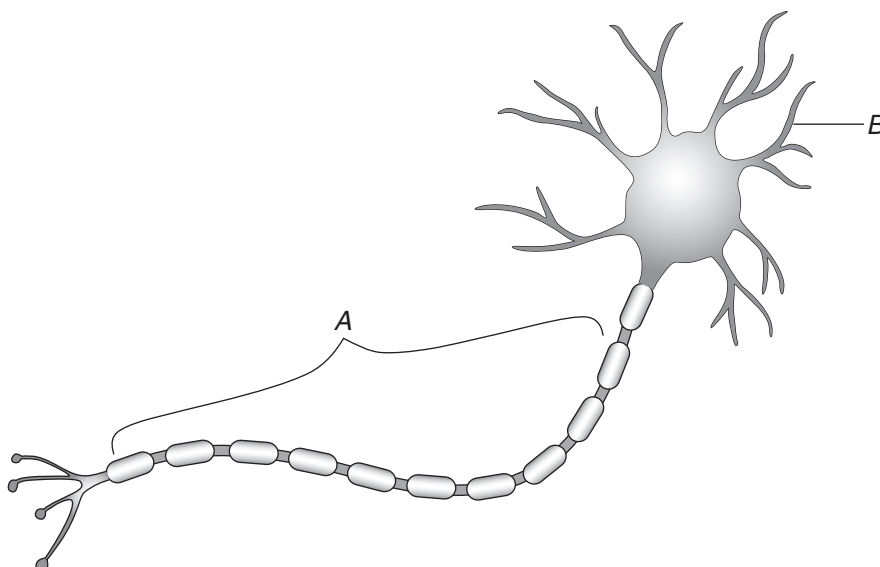
Extra writing booklets are available.

	Pages
Question 32 Communication	26–27
Question 33 Biotechnology	28–29
Question 34 Genetics: The Code Broken?	30–31
Question 35 The Human Story	32–33
Question 36 Biochemistry	34–35

Question 32 — Communication (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The diagram below shows a microscopic view of a neuron.



- | | | |
|---|--|----------|
| (i) | What are the structures labelled <i>A</i> and <i>B</i> ? Clearly indicate in your answer which is <i>A</i> and which is <i>B</i> . | 2 |
| (ii) | Identify the role of structure <i>B</i> . | 1 |
|
(b) Explain why some stimuli do not result in an action potential. Use a graph to support your answer. | | 4 |
|
(c) Laser Assisted In-Situ Keratomileusis (LASIK) is a procedure that uses a laser beam to cut into the cornea to change its shape in a precise manner. | | |
| (i) | Identify the functions of the cornea. | 2 |
| (ii) | Explain how LASIK could be used to treat myopia so that glasses or contact lenses are no longer required. | 3 |

Question 32 continues on page 27

Question 32 (continued)

Answer parts (d)–(e) in Section II Answer Booklet 2.

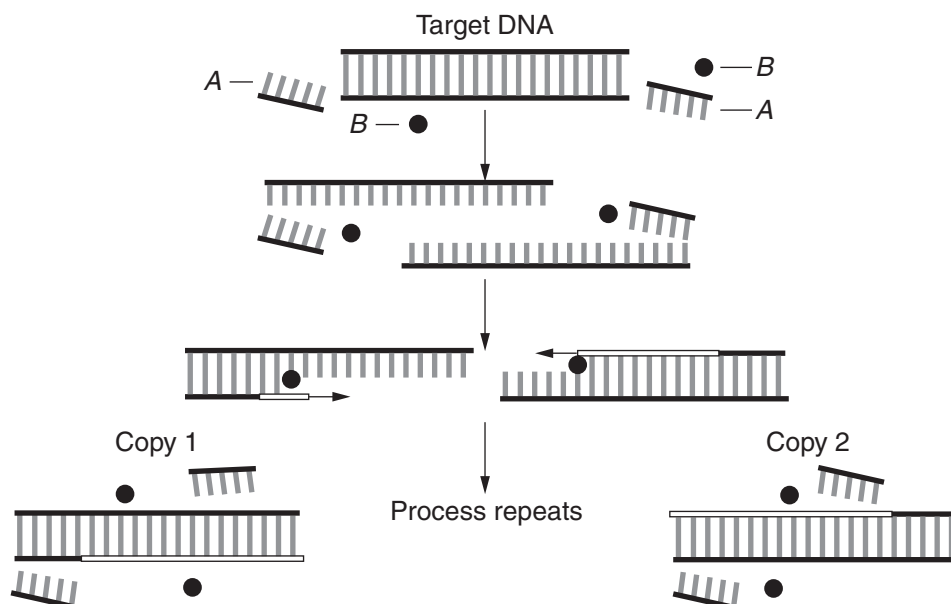
- | | | |
|-----|--|----------|
| (d) | Compare the hearing of THREE mammals, including humans, giving possible reasons for their differences in hearing. | 6 |
| (e) | Explain how the hearing aid and cochlear implant technologies assist hearing. Refer to the anatomy and function of the human ear and the role of the brain in coordinating responses to stimuli. | 7 |

End of Question 32

Question 33 — Biotechnology (25 marks)

Answer parts (a)–(c) in the Section II Answer Booklet 1.

- (a) The diagram below shows a recombinant DNA technique.



- (i) What are represented by the components labelled *A* and *B*? Clearly indicate in your answer which is *A* and which is *B*. 2
- (ii) Identify the process shown in the diagram. 1
- (b) Explain why transcription and translation result in different products. Use a diagram to support your answer. 4

- (c)

Ants cultivate grass for food

A species of ant has been observed to clear the surface of the ground for 5 metres around its colony and bury grass seeds there. The ants remove any other plants that may spring up among their crop. When the seeds are ripe, the ants harvest the seeds for food and retain some seeds for next year's crop. A ten-year study has shown that the average size of the grass seeds increased by 5%.

- (i) Outline the difference between qualitative and quantitative observations, using examples from the text above. 2
- (ii) Explain why the ant activity could be interpreted as biotechnology. 3

Question 33 continues on page 29

Question 33 (continued)

Answer parts (d)–(e) of the question in Section II Answer Booklet 2.

- (d) Explain the risks and benefits to society of using recombinant DNA technologies. **6**
- (e) Enzymes have been developed from bacteria that remove contaminating insecticides and herbicides from waterways. **7**

Explain how the technologies you have studied in this option could be applied to the development and production of such enzymes. Refer to techniques such as those used to extract enzymes, amplify genes and recombine DNA.

End of Question 33

Question 34 — Genetics: The Code Broken? (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

(a) The table shows parts of DNA sequences.

	DNA Sequence
<i>Original DNA</i>	AAC TCG GTC AAT ATG
<i>Mutation 1</i>	AAC TCC GTC AAT ATG
<i>Mutation 2</i>	AAC TCG GTA ATA TGC

- (i) What types of mutations are *Mutation 1* and *Mutation 2*? Clearly indicate in your answer which is *Mutation 1* and which is *Mutation 2*. 2
- (ii) Using the information provided in the table, what is the effect of *Mutation 1* in part (i)? 1

	T	C	A	G
T	TTT Phe (F) TTC Phe (F) TTA Leu (L) TTG Leu (L)	TCT Ser (S) TCC Ser (S) TCA Ser (S) TCG Ser (S)	TAT Tyr (Y) TAC Tyr (Y) TAA STOP TAG STOP	TGT Cys (C) TGC Cys (C) TGA STOP TGG Trp (W)
C	CTT Leu (L) CTC Leu (L) CTA Leu (L) CTG Leu (L)	CCT Pro (P) CCC Pro (P) CCA Pro (P) CCG Pro (P)	CAT His (H) CAC His (H) CAA Gin (Q) CAG Gin (Q)	CGT Arg (R) CGC Arg (R) CGA Arg (R) CGG Arg (R)
A	ATT Ile (I) ATC Ile (I) ATA Ile (I) ATG Met (M) START	ACT Thr (T) ACC Thr (T) ACA Thr (T) ACG Thr (T)	AAT Asn (N) AAC Asn (N) AAA Lys (K) AAG Lys (K)	AGT Ser (S) AGC Ser (S) AGA Arg (R) AGG Arg (R)
G	GTT Val (V) GTC Val (V) GTA Val (V) GTG Val (V)	GCT Ala (A) GCC Ala (A) GCA Ala (A) GCG Ala (A)	GAT Asp (D) GAC Asp (D) GAA Glu (E) GAG Glu (E)	GGT Gly (G) GGC Gly (G) GGA Gly (G) GGG Gly (G)

Question 34 continues on page 31

Question 34 (continued)

- (b) Explain how children with different ABO and Rhesus blood groups could be the offspring of the same parents. Use diagrams to support your answer. **4**

- (c) Generally, chickens raised for meat are fed intensively to achieve fast growth. However, this also leads to the generation of a lot of body heat.

It has been found that appetite is reduced when animals are hot. A scientist, using the technology of selective breeding, has developed a featherless chicken.

The scientist claims that ‘these chickens are not genetically modified as they come from a natural breed and were not cloned’.

- (i) Outline the difference between selective breeding and animal cloning. **2**

- (ii) Discuss the scientist’s claim using your knowledge of selective breeding. **3**

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) Discuss the use of the Human Genome Project and traditional inheritance studies as ways to identify the location of harmful genes. **6**

- (e) The origin of the dingo (an Australian native dog) is uncertain. It has commonly been classified as *Canis familiaris dingo*, a subspecies of the domestic dog, *Canis familiaris*. **7**

It has recently been suggested that the dingo should be reclassified as *Canis lupus dingo* to reflect a closer relationship to the Asian wolf, *Canis lupus*.

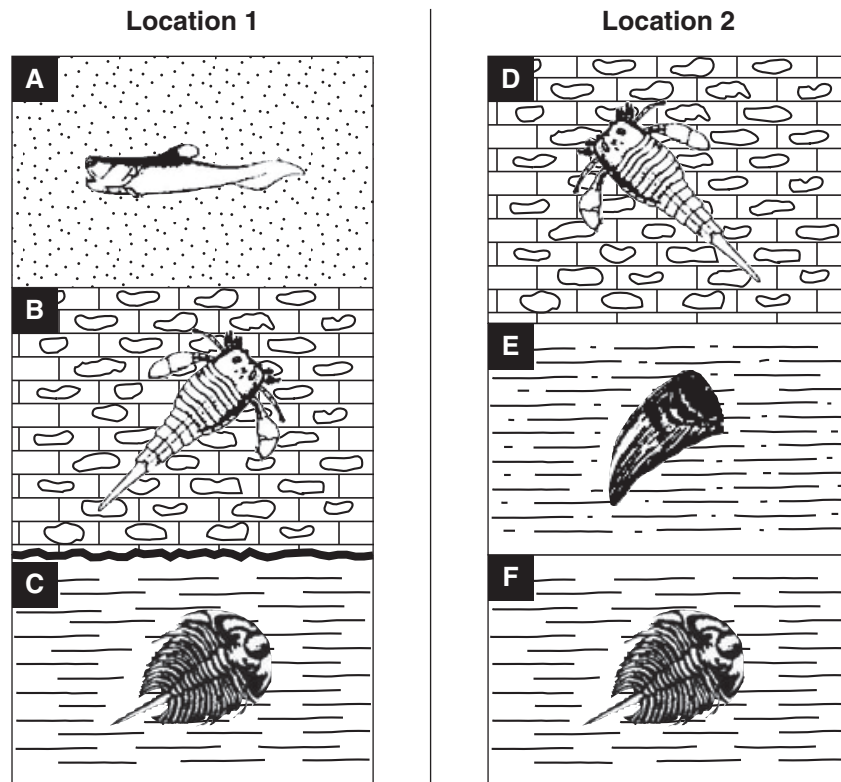
Explain how the technologies you have studied in this option could be applied to identifying the relationships between these species.

End of Question 34

Question 35 — The Human Story (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

- (a) The diagrams show labelled fossil strata that were found in different locations within a region.



- (i) Which rock layer(s) is/are the oldest and which is/are the youngest in this region? 2
- (ii) Name the relative dating technique used in your answer to part (i). 1
- (b) Explain TWO alternative views of the evolutionary relationships between FOUR identified hominids of the same genus. Use diagrams to support your answer. 4

Question 35 continues on page 33

Question 35 (continued)

- (c) In 2004, a new species, *Homo floresiensis*, was identified from fossils found in a cave on an Indonesian island. Dated to 18 000 years ago, these fossils were found in sediment containing stone tools and cooking hearths.

The adult skull of this upright bi-pedal fossil organism has a volume of 380 cm³. The size of each adult organism was approximately the same as a three-year-old modern human.

Their arm to leg ratio was slightly larger than modern humans. They had hard, thicker eyebrow ridges than us, a sharply sloping forehead, and no chin.

The cooking hearths contained charred bones of game species, each estimated to weigh more than 1 000 kilograms. Stone tools included blades, spear heads and cutting and chopping utensils.

- | | |
|---|----------|
| (i) Outline the fossil evidence from this description that suggests the advanced cultural development of <i>Homo floresiensis</i> . | 2 |
| (ii) Discuss the validity of the classification of <i>Homo floresiensis</i> as a species separate from other known fossil hominids. | 3 |

Answer parts (d)–(e) in Section II Answer Booklet 2.

- | | | |
|-----|--|----------|
| (d) | (i) By explaining the mechanisms of biological evolution, compare early and current <i>Homo sapiens</i> . | 4 |
| | (ii) Explain why there is a greater genetic diversity between human populations in Africa than between human populations in other continents. | 2 |
| (e) | Insectivorous microbats and fruit-eating megabats are classified as members of the order Chiroptera. Lemurs are fruit and leaf eaters and are classified as members of the order Primates. | 7 |

Scientists have recently hypothesised that megabats are closer in evolutionary terms to lemurs than to microbats.

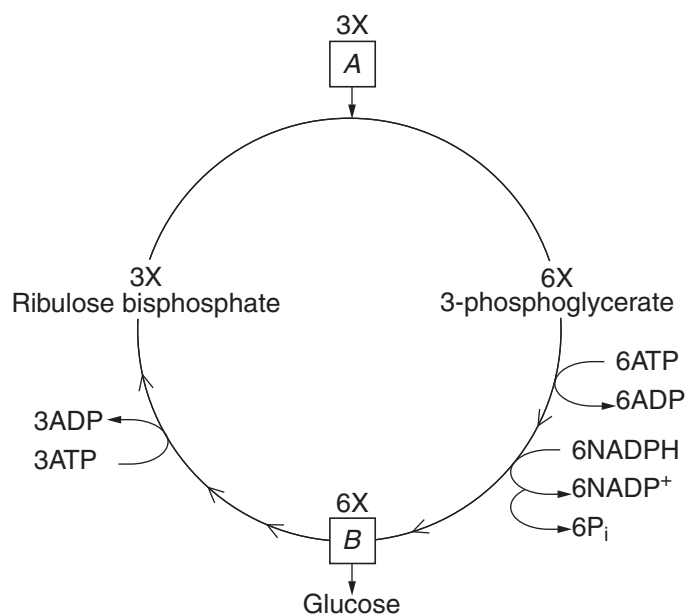
Explain how you could use technologies and the data they generate to test this hypothesis.

End of Question 35

Question 36 — Biochemistry (25 marks)

Answer parts (a)–(c) in Section II Answer Booklet 1.

The diagram outlines a biochemical pathway.



- (a) (i) Name the pathway. 1
- (ii) What are molecules *A* and *B*? Clearly indicate in your answer which is *A* and which is *B*. 2
- (b) Describe the location of the site of light absorption during photosynthesis. Use a diagram to support your answer. 4

Question 36 continues on page 35

Question 36 (continued)

- (c) You wish to study the dark cycle of photosynthesis by using radioactive isotopes. A table of half-lives is provided. **5**

Element	Isotope	Half-life
Carbon	C-11	20.38 minutes
	C-14	5730 years
Hydrogen	H-3	12.35 years
Nitrogen	N-13	9.97 minutes
	N-16	7.13 seconds
Oxygen	O-15	122.24 seconds
Sulfur	S-35	87.44 days

Name the isotopes you would use for this study and account for your choice.

Answer parts (d)–(e) in Section II Answer Booklet 2.

- (d) Describe how Mayer's conclusion about photosynthesis brought together the ideas from the work of earlier scientists. **6**

- (e) A new chlorophyll, chlorophyll E, was recently discovered in cyanobacteria by Australian scientists. As a member of the scientific team, your role is to identify the subcellular location of the new chlorophyll, extract it, separate it from other chlorophylls and analyse its properties. **7**

Explain the technologies that you would use to carry out your role in the scientific team.

End of paper

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2011 HSC Biology Marking Guidelines

Section I, Part A

Multiple-choice Answer Key

Question	Answer
1	B
2	C
3	B
4	C
5	B
6	D
7	B
8	B or D
9	C
10	D
11	A
12	B
13	A
14	B
15	B
16	A
17	D
18	A
19	A
20	C

Section I, Part B

Question 21 (a)

Criteria	Marks
• Names TWO methods used for treating drinking water to reduce the risk of infection	2
• Names a method used for treating drinking water to reduce the risk of infection	1

Question 21 (b)

Criteria	Marks
• Sketches in general terms how each of the named methods reduce the risk of infection	2
• Sketches in general terms how one named method reduces the risk of infection	1

Question 22 (a)

Criteria	Marks
• Correctly states the purpose of the experiment	1

Question 22 (b)

Criteria	Marks
• Correctly provides characteristic and/or features of TWO improvements that relate specifically to the validity of the data collected in the experiment	2
• Correctly provides characteristic and/or feature of ONE improvement that relates specifically to the validity of the data collected in the experiment	1

Question 22 (c)

Criteria	Marks
• Correctly predicts the pHs that result in the production of clear and cloudy solutions	1

Question 23

Criteria	Marks
- Names at least THREE scientists involved in determining the structure of DNA - Gives a thorough description of the collaboration and communication between these scientists - Clearly relates this to the effect on their progress in determining the structure of DNA	5
- Names at least TWO scientists involved in determining the structure of DNA - Gives a thorough description of the collaboration and communication between some of these OR - Relates it to their ideas about the structure of DNA	4
- Names at least TWO scientists involved in determining the structure of DNA - Gives a description of the collaboration and communication between some of these OR - Gives a thorough description of the collaboration and communication between non-named but identified scientists - Clearly relates this to the effect on their progress to determining the structure of DNA	3
- Names some scientists involved in determining the structure of DNA AND EITHER - Gives a description of the collaboration and communication between some of these OR - Identifies their roles in the discovery of DNA	2
- Names at least ONE scientist involved in determining the structure of DNA OR - Provides a relevant statement about the impact of collaboration or communication on scientific research OR - Provides some correct information about the structure of DNA	1

Question 24 (a)

Criteria	Marks
Identifies a viral disease that is controlled by the use of a vaccine	1

Question 24 (b) (i)

Criteria	Marks
Indicates ONE main feature of the viral vaccine and links it to a named immune response	2
- Indicates ONE main feature of a vaccine OR - Names an immune response	1

Question 24 (b) (ii)

Criteria	Marks
Identifies a correct step from the flowchart and relates it to safe use of the vaccine	2
- Identifies a correct step from the flowchart OR - Relates it to a safe use of the vaccine	1

Question 25 (a)

Criteria	Marks
Identifies BOTH processes that takes place at location identified	2
Identifies ONE process that takes place at location identified	1

Question 25 (b)

Criteria	Marks
Identifies the process that occurs due to active transport	1

Question 25 (c)

Criteria	Marks
- Sketches in general terms the increase of aldosterone leading to an increase in salt - Water retention in blood	2
- Sketches in general terms the action of aldosterone in the regulation of salt OR - Water retention in blood	1

Question 26 (a)

Criteria	Marks
• Provides TWO differences and TWO similarities in the data	4
• Provides THREE of the above	3
• Provides TWO of the above	2
• Provides ONE of the above	1

Question 26 (b)

Criteria	Marks
• Provides reasons in favour of TWO additional types of data required	2
• Provides a reason in favour of ONE additional type of data required OR • Identifies TWO additional data types	1

Question 27

Criteria	Marks
• Sketches in general terms FOUR features of an epidemiological study to identify the cause of the disease	4
• Sketches in general terms features of an epidemiological study to identify the cause of the disease	3
• Describes a feature of an epidemiological study to identify the cause of the disease	2
• Identifies a feature of an epidemiological study	1

Question 28 (a)

Criteria	Marks
• Correctly indicates the activation of stem cells • Relates activation to restoration of the skin barrier	2
• Correctly indicates the activation of stem cells OR • Relates activation to restoration of the skin barrier	1

Question 28 (b)

Criteria	Marks
Provides TWO features or characteristics that might be an improvement and relates ONE to the performance of an artificial blood	3
- Provides TWO features or characteristics that might be an improvement OR - Provides ONE feature or characteristic that might be an improvement and relates it to the performance	2
Provides ONE feature or characteristic that might be an improvement	1

Question 29

Criteria	Marks
- Completes the table by: - identifying dependent variable - identifying independent variable - identifying a control - providing TWO risks related to safety precaution	5
Completes the table by completing FOUR aspects of the investigation	4
Completes the table by completing THREE aspects of the investigation	3
Completes the table by completing TWO aspects of the investigation	2
Completes the table by completing ONE aspect of the investigation	1

Question 30 (a)

Criteria	Marks
- Names technology X - Names technology Y	2
- Names technology X OR - Names technology Y	1

Question 30 (b)

Criteria	Marks
- Provides a correct statement about the conclusion being invalid - Builds a sequence of correct ideas to justify this view based on the experiment	3
- Provides a correct statement about the conclusion being invalid - Gives a correct idea to justify this view based on this experiment OR - Builds a sequence of correct ideas to justify this view based on the experiment	2
- Provides a correct statement about the conclusion being invalid OR - Gives a correct idea to justify this view based on this experiment OR - Provides a correct statement about reptiles controlling their body temperature	1

Question 31

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of advances in our understanding of biology and their implications for society • Gives a thorough description of major advances in biology that have allowed the development and use of antibiotics • Relates the implications of the development of antibiotics to society • Demonstrates coherence and logical progression and includes correct use of biological terms	6–7
• Demonstrates a sound understanding of advances in our understanding of biology and their implications for society • Gives a description of TWO major advances in biology that have allowed the development and use of antibiotics • Outlines the implications of the development of antibiotics to society • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates some understanding of advances in our understanding of biology and their implications for society • Gives a description of ONE major advance in our understanding that has allowed the development and use of antibiotics • Outlines an implication of the development of antibiotics to society • Communicates ideas in a basic form using some general scientific terms	2–3
• Provides a relevant statement about a major advance in our understanding of biology that could be linked to the development of antibiotics OR • Outlines an implication of the development of antibiotics to society	1

Section II

Question 32 (a) (i)

Criteria	Marks
• Identifies structures <i>A</i> and <i>B</i>	2
• Identifies either structure <i>A</i> or structure <i>B</i>	1

Question 32 (a) (ii)

Criteria	Marks
• Names the role of the dendrite	1

Question 32 (b)

Criteria	Marks
• Draws a graph that correctly shows the relationship between stimulus and changing membrane potential and indicates threshold • Relates the level of stimulus to the formation (or not) of an action potential	4
• Draws a graph that correctly shows the relationship between stimulus and changing membrane potential and indicates threshold OR • Draws a graph that correctly shows the relationship between stimulus and changing membrane potential and relates the level of stimulus to the formation (or not) of an action potential OR • Draws a graph that correctly shows changing membrane potential and indicates threshold, and relates the level of stimulus to the formation (or not) of an action potential	3
• Draws a diagram or graph that correctly shows the relationship between stimulus and changing membrane potential OR • Draws a diagram or graph that correctly shows changing membrane potential and indicates threshold OR • Draws a diagram that correctly shows a stimulus and indicates threshold OR • Relates the level of stimulus to the formation (or not) of an action potential	2
• Draws a diagram or graph that indicates some information about membrane potential and stimulus OR • Relates some information about a stimulus and action potential	1

Question 32 (c) (i)

Criteria	Marks
• Names at least TWO roles of the cornea	2
• Names a role of the cornea	1

Question 32 (c) (ii)

Criteria	Marks
- Correctly identifies the problem associated with myopia - Provides correct information on how LASIK corrects this defect	3
- Correctly identifies the problem associated with myopia - Identifies that the surgery is used to correct focus	2
- Provides some correct information about myopia OR - Identifies that the surgery is used to correct focus	1

Question 32 (d)

Criteria	Marks
- Compares the range of frequencies detected by humans and TWO other mammals - Relates possible reasons for differences between the range of frequencies detected by humans and TWO other mammals	5–6
- Identifies the range of relative frequencies detected by humans and ONE other mammal - Relates possible reasons for differences between the range of relative frequencies detected by humans and ONE other mammal OR - Describes the structure of a human ear and compares it to the different structure used by ONE other mammal - Relates possible reasons for this different structural arrangement	3–4
- Identifies a correct relative range of frequencies detected by humans and ONE other mammal OR - Identifies a possible reason for differences between the range of relative frequencies detected by TWO mammals OR - Identifies a reason for differences in the anatomical arrangement of the ear structure in a human and ONE other mammal	2
- Identifies the range of frequencies detected by a mammal OR - Identifies a reason for a mammal to hear within a certain frequency range OR - Identifies differences in the ear structure in ONE other mammal	1

Question 32 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the anatomy and function of the human ear and the role of the brain in responding to stimuli • Links how the use of hearing aids and cochlear implants assists hearing with reference to the conditions • Demonstrates coherence and logical progression and includes correct use of biological terms	6–7
• Demonstrates a sound understanding of the anatomy and function of the human ear and the role of the brain in responding to stimuli • Outlines how the use of hearing aids and cochlear implants assists hearing with reference to at least one condition • Communicates some scientific principles and ideas in a clear manner OR • Demonstrates a thorough knowledge and understanding of the anatomy and function of the human ear to send messages to the brain for interpretation OR • Outlines how the use of hearing aids and cochlear implants assist hearing with reference to at least one condition • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates some understanding of the anatomy and function of the human ear and/or the role of the brain in responding to aural stimuli • Identifies the use of hearing aids or cochlear implants in assisting hearing • Communicates ideas in a basic form using some general scientific terms	2–3
• Identifies the use of hearing aids or cochlear implants in assisting hearing OR • Identifies the role of the brain in coordinating the correct interpretation of sensory signals	1

Question 33 (a) (i)

Criteria	Marks
• Identifies components A and B	2
• Identifies component A or B	1

Question 33 (a) (ii)

Criteria	Marks
• Names the process shown in the diagram	1

Question 33 (b)

Criteria	Marks
- Draws a diagram that correctly depicts transcription and translation - Relates the production of different products to transcription and translation	4
- Draws a diagram that correctly depicts transcription and translation OR - Relates the production of different products to transcription and translation	3
Any TWO of the following: - Draws a diagram that correctly depicts transcription - Relates the production of different products to transcription - Draws a diagram that correctly depicts translation - Relates the production of different products to translation	2
- Draws a diagram that depicts some information about transcription OR - Draws a diagram that depicts some information about translation OR - Relates the production of different products to transcription or translation	1

Question 33 (c) (i)

Criteria	Marks
Distinguishes between qualitative and quantitative data and gives examples of each taken from the text	2
- Distinguishes between qualitative and quantitative data without examples from the text OR - Gives examples from the text but does not distinguish between quantitative and qualitative data	1

Question 33 (c) (ii)

Criteria	Marks
Correctly defines biotechnology and provides an argument for or against the activity of the ants being considered as biotechnology and provides the outcome	3
- Correctly defines biotechnology and provides an argument for or against the activity of the ants being considered as biotechnology OR - Provides an argument for or against the activity of the ants being considered as biotechnology and provides the outcome OR - Correctly defines biotechnology and provides the outcome	2
- Correctly defines biotechnology OR - Identifies the activity of the ants being considered as biotechnology OR - Provides the outcome of the ants' activity	1

Question 33 (d)

Criteria	Marks
- Defines recombinant DNA technology - Indicates the purpose of recombinant DNA technology - Shows the relationship between the use of recombinant DNA technologies and risks AND benefits to society	5–6
- Indicates the outcome of recombinant DNA technologies - Shows the relationship between the use of recombinant DNA technologies and the risks OR benefits to society	3–4
- Identifies risks OR benefits to society OR - Identifies ONE risk and ONE benefit to society OR - Shows the relationship between the use of recombinant DNA technologies and ONE risk OR benefit to society	2
Identifies a risk OR benefit to society	1

Question 33 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of technologies that could be used to extract enzymes, amplify genes or recombine DNA • Correctly identifies features and characteristics of THREE technologies that could be applied to developing or producing the enzymes • Clearly states how the technologies are linked to development or improved production • Demonstrates coherence and logical progression and includes correct use of biological terms	6–7
• Demonstrates sound understanding of technologies that could be used to extract enzymes, amplify genes or recombine DNA • Correctly identifies features and characteristics of TWO technologies that could be applied to developing or producing the enzymes • Clearly states how the technologies are linked to development or improved production OR • Correctly identifies THREE technologies that could be applied to developing or producing the enzymes • Clearly states how a technology is linked to development or improved production • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates some understanding of technologies that could be used to extract enzymes, amplify genes or recombine DNA • Correctly identifies TWO technologies that could be applied to developing or producing the enzymes • Clearly states how the technologies are linked to development or improved production OR • Correctly identifies THREE technologies that could be applied to developing or producing the enzymes • Communicates ideas in a basic form using some general scientific terms	3
• Demonstrates an understanding of technologies that could be used to extract enzymes, amplify genes or recombine DNA • Correctly identifies TWO technologies that could be applied to developing or producing the enzymes OR • Correctly identifies ONE technology that could be applied to developing or producing the enzymes • States how the technologies are linked to development or improved production	2
• Correctly identifies and describes a technology that could be applied to developing or producing enzymes	1

Question 34 (a) (i)

Criteria	Marks
• Names types of mutations for 1 and 2	2
• Names types of mutations for either 1 or 2	1

Question 34 (a) (ii)

Criteria	Marks
• Identifies the effect of the mutation	1

Question 34 (b)

Criteria	Marks
• Draws Punnett squares or other diagram that correctly show inheritance of ABO blood groups and Rhesus factors to produce offspring with different blood group genotypes and phenotypes from the parent genotypes	4
• Draws Punnett squares or other diagram that correctly show inheritance of ABO blood groups and Rhesus factors to produce offspring with different blood group genotypes from the parent genotypes	3
• Draws a Punnett square or clearly indicates inheritance of ABO blood groups or Rhesus factors to produce offspring with different blood group genotypes and/or phenotypes from the parent genotypes	2
• Uses correct genotype notation for either ABO blood group or Rhesus factor	1

Question 34 (c) (i)

Criteria	Marks
• Provides main features of selective breeding • Provides main features of cloning	2
• Provides main features of selective breeding OR • Provides main features of cloning OR • Provides ONE feature of both selective breeding and cloning	1

Question 34 (c) (ii)

Criteria	Marks
- Identifies parts in favour of and against the scientific claim - Indicates main features of both selective breeding and cloning with regard to the change in genetic nature in a species	3
- Identifies parts in favour of and against the scientific claim - Indicates main features of selective breeding with regard to the change in genetic nature in a species OR - Identifies parts in favour of and against the scientific claim - Indicates main features of cloning with regard to the change in genetic nature in a species	2
- Makes some correct statement about selective breeding OR - Makes some correct statement about cloning OR - Makes some correct statement about genetic modification	1

Question 34 (d)

Criteria	Marks
Clearly identifies the type of information and methods used to acquire information in the HGP and traditional inheritance studies	5–6
Clearly identifies the type of information in the HGP and traditional inheritance studies	3–4
- Makes a correct statement about the way the HGP is used to identify the location of genes - Makes a correct statement about the way traditional inheritance studies are used to identify the location of genes	2
- Makes a correct statement about the way the HGP is used to identify the location of genes OR - Makes a correct statement about the way traditional inheritance studies are used to identify the location of genes	1

Question 34 (e)

Criteria	Marks
- Provides features and characteristics of THREE technologies that could be applied to study relationships between species - Demonstrates a thorough knowledge and understanding of technologies that could be applied to identifying the relationships - Clearly states how the technologies are linked to the example of dingo relationships - Demonstrates coherence and logical progression and includes correct use of biological terms	6–7
- Provides features and characteristics of TWO technologies that could be applied to study relationships between species - Demonstrates a sound understanding of technologies that could be applied to identifying the relationships - Clearly states how the technologies are linked to the example of dingo relationships OR - Correctly identifies THREE technologies that could be applied to study relationships between species and a characteristic or feature of ONE technology - Clearly states how the technologies are linked to the example of dingo relationships - Communicates some scientific principles and ideas in a clear manner	4–5
- Correctly identifies TWO technologies that could be applied to study relationships between species and a characteristic or feature of ONE technology - Demonstrates some understanding of technologies that could be applied to identifying the relationships - Clearly states how the technologies are linked to the example of dingo relationships OR - Correctly identifies THREE technologies that could be applied to study relationships between species - Clearly states how the technologies are linked to the example of dingo relationships - Communicates ideas in a basic form using some general scientific terms	3
- Correctly identifies THREE technologies that could be applied to study relationships between species - Demonstrates a limited understanding of technologies that could be applied to identifying the relationships OR - Correctly identifies TWO technologies that could be applied to study relationships between species and describes ONE technology OR	2

• Correctly identifies TWO technologies that could be applied to study relationships between species • Clearly states how the technologies are linked to the example of dingo relationships	
• Correctly identifies TWO technologies that could be applied to study relationships between species OR • Correctly identifies and describes ONE technology that could be applied to study relationships between species	1

Question 35 (a) (i)

Criteria	Marks
• Identifies the oldest layer twice and the youngest layer	2
• Identifies the oldest layer twice or the youngest layer	1

Question 35 (a) (ii)

Criteria	Marks
• Identifies the relative dating technique	1

Question 35 (b)

Criteria	Marks
• Draws diagrams that correctly support TWO alternate views of the evolutionary relationship between FOUR identified hominids of the same genus	4
• Draws diagrams that correctly support TWO alternative views of the evolutionary relationships between TWO or THREE hominids of the same genus	3
• Correctly identifies TWO alternative views of the evolutionary relationship between at least TWO identified hominids	2
• Correctly identifies some information about hominid evolution	1

Question 35 (c) (i)

Criteria	Marks
• Links TWO appropriate pieces of data from source to evidence of advanced culture	2
• Links ONE appropriate piece of data from source to evidence of advanced culture OR • Identifies TWO appropriate pieces of data	1

Question 35 (c) (ii)

Criteria	Marks
- Provides a correct definition of species - Identifies the unique combination of hominid features in <i>Homo floresiensis</i> - Identifies isolation and space and time as appropriate for speciation OR - Provides arguments for and/or against validity - Identifies the unique combination of hominid features in <i>Homo floresiensis</i> - Identifies isolation and space and time as appropriate speciation	3
TWO of the above	2
ONE of the above	1

Question 35 (d) (i)

Criteria	Marks
- Provides information about mechanisms of evolution - Correctly identifies similarities AND/OR differences between the evolution of early and current <i>Homo sapiens</i>	3–4
- Provides information about mechanisms of evolution - Identifies some characteristics of the evolution of both early and current <i>Homo sapiens</i>	2
- Provides some information about mechanisms of evolution OR - Identifies some characteristics of the evolution of both early and current <i>Homo sapiens</i>	1

Question 35 (d) (ii)

Criteria	Marks
- Relates increased time for evolution in Africa to longer opportunity for divergence - Relates recent migration from some parts of Africa to new isolated habitats to shorter opportunity for divergence	2
- Relates increased time for evolution in Africa to longer opportunity for divergence OR - Relates recent migration from some parts of Africa to new isolated habitats to shorter opportunity for divergence	1

Question 35 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of technologies and data that could be used to test the hypothesis • Provides characteristics and features of TWO technologies • Identifies the type of data that can be obtained from TWO technologies • Identifies how data is analysed for each of the TWO technologies • Relates the data analysis to the specific matter of the degree of evolutionary proximity between the TWO species • Demonstrates coherence and logical progression and includes correct use of biological terms	6–7
• Demonstrates sound understanding of technologies and data that could be used to test the hypothesis • Provides characteristics and features of TWO technologies • Identifies the type of data that can be obtained from the TWO technologies • Identifies how data are analysed for each of the TWO technologies • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates some understanding of technologies and data that could be used to test the hypothesis • Provides characteristics and features of TWO technologies • Identifies the type of data that can be obtained from TWO technologies (same technologies) • Communicates ideas in a basic form using some general scientific terms	3
• Demonstrates a limited understanding of technologies and data that could be used to test the hypothesis • Provides characteristics and features of TWO technologies OR • Identifies the type of data that can be obtained from TWO technologies identified OR • Identifies how data are analysed for each of TWO technologies OR • Identifies a number of relevant features for ONE technology	2
• Identifies ONE appropriate technology	1

Question 36 (a) (i)

Criteria	Marks
• Names the pathway	1

Question 36 (a) (ii)

Criteria	Marks
• Names both molecules <i>A</i> and <i>B</i>	2
• Names either molecule <i>A</i> or <i>B</i>	1

Question 36 (b)

Criteria	Marks
• Draws a diagram showing the main sites of light absorption and provides characteristics and features of the photosystem	4
• Draws a diagram showing the main sites of light absorption and provides a characteristic or feature of a photosystem	3
• Draws a diagram OR clearly indicates some characteristics related to photosynthesis	2
• Provides some information about light absorption and photosynthesis	1

Question 36 (c)

Criteria	Marks
• States the correct reasons for AND/OR against using the isotopes	4–5
• States most of the reasons for AND/OR against using the isotope	2–3
• States a reason for OR a reason against using the isotope	1

Question 36 (d)

Criteria	Marks
• Provides detailed characteristics and features of how the work of the other named scientists was brought together by Mayer's ideas	5–6
• Provides some characteristics and features of how the work of at least ONE named scientist was brought together by Mayer's ideas	3–4
• Provides some information about Mayer's ideas on photosynthesis	2
• Provides some information about photosynthesis correctly linked to a scientist	1

Question 36 (e)

Criteria	Marks
• Demonstrates a thorough knowledge and understanding of the technologies that would be used to carry out your role in the scientific team • Provides characteristics and features of TWO technologies • Identifies the type of data that can be obtained from the TWO technologies • Identifies how data is analysed for each of the TWO technologies • Relates the use of the technology to testing the hypothesis • Demonstrates coherence and logical progression and includes correct use of biological terms	6–7
• Demonstrates a sound understanding of the technologies that would be used to carry out your role in the scientific team • Provides characteristics and features of TWO technologies • Identifies the type of data that can be obtained from the TWO technologies • Identifies how the data is analysed for each of the technologies • Communicates some scientific principles and ideas in a clear manner	4–5
• Demonstrates some understanding of the technologies that would be used to carry out your role in the scientific team • Provides characteristics and features of TWO technologies • Identifies the type of data that can be obtained from the TWO technologies • Communicates ideas in a basic form using some general scientific terms	3
• Demonstrates a limited understanding of the technologies that would be used to carry out your role in the scientific team • Provides characteristics and features of TWO technologies OR • Identifies the type of data that can be obtained from the TWO technologies OR • Identifies how data are analysed for each technology	2
• Identifies ONE appropriate technology	1

Biology

2011 HSC Examination Mapping Grid

Section I Part A

Question	Marks	Content	Syllabus outcomes
1	1	9.3.1.2.2, 9.3.3.2.2	H6, H9
2	1	9.2.2.2.1	H6
3	1	9.2.1.3.1, 9.2.1.2.3, 9.2.1.2.4, 9.2.1.2.6	H6
4	1	9.3.3.2.2, 9.3.3.2.3	H9
5	1	9.3.2.2.6, 9.3.2.2.3, 9.3.2.2.4, 9.3.2.3.2	H9
6	1	9.4.5.1	H1, H3
7	1	9.4.2.2.2, 9.4.2.1	H4
8	1	9.4.5.2.1, 9.4.5.2.2	H6
9	1	9.4.3.2.2, 14.3c	H6, H14
10	1	9.3.4.2.5, 9.3.5.2.1, 9.3.3.2.5	H7, H9
11	1	9.2.2.2.6, 14.1a, 14.1g, 14.1f, 12.3c	H6, H12
12	1	9.2.3.3.5	H7
13	1	9.2.2.2.4, 12.3c	H6, H12
14	1	9.4.4.2.1	H6
15	1	9.3.1.2.2, 14.1g	H10, H14
16	1	9.4.5.2.1, 9.4.5.2.2, 12.3c	H6, H12
17	1	9.3.2.2.4, 9.3.3.3.2, 14.1a, 14.1f	H9, H14
18	1	9.2.2.2.3, 12.3c, 14.1a	H6, H12, H14
19	1	9.4.3.3.2, 9.4.3.2.3	H8
20	1	9.3.4.2.7	H10

Part B

Question	Marks	Content	Syllabus outcomes
21 (a)	2	9.4.2.3.2	H3, H4, H6
21 (b)	2	9.4.2.3.2	H3, H4, H6
22 (a)	1	9.2.1.3.1, 9.2.1.2.1, 9.2.1.2.2, 14.1a, 12.3c	H14, H12
22 (b)	2	9.2.1.3.1, 14.3c	H14
22 (c)	1	9.2.1.3.1, 14.1c	H14
23	5	9.3.4.3.5	H1, H2
24 (a)	1	9.4.5.3.1	H3, H4, H8
24 (b) (i)	2	9.4.5.3.1	H3, H4, H6
24 (b) (ii)	2	9.4.5.2.3	H3, H4, H6
25 (a)	2	9.2.3.2.6, 14.1f	H6, H14
25 (b)	1	9.2.3.2.5	H6
25 (c)	2	9.2.3.2.7	H6
26 (a)	4	9.4.6, 12.3c	H12

26 (b)	2	9.4.6, 12.4a	H12
27	4	9.4.6.1, 11.2c	H11
28 (a)	2	9.4.1.2.2, 9.4.1.3.1, 9.4.4.2.1, 12.3c	H12
28 (b)	3	9.2.2.3.5, 12.3c	H3, H12
29	5	9.4.2.3.1, 11.2a	H11
30 (a)	2	11.1e, 12.2a	H11, H12
30 (b)	3	9.2.1.3.3, 9.2.1.2.8, 14.1b	H6, H14
31	7	9.2.2.1, 9.2.1.3.1, 9.3.3.2.2, 9.3.4.2.3, 9.3.4.2.4, 9.3.4.2.5, 9.4.2.2.1, 9.4.2.3, 9.4.3, 9.4.3.2.2, 14.3b	H1, H4, H8, H7, H10, H14

Section II

Question 32 — Communication

Question	Marks	Content	Syllabus outcomes
32 (a) (i)	2	9.5.7.3.1	H6, H13
32 (a) (ii)	1	9.5.7.3.1	H6
32 (b)	4	9.5.7.2.3, 9.5.7.3.3, 13.1b, 13.1e, 13.1f	H6, H13
32 (c) (i)	2	9.5.2.2.1, 9.5.3.3.1, 9.5.3.2.2	H6
32 (c) (ii)	3	9.5.3.2.5	H6, H4
32 (d)	6	9.5.6.3.2	H6
32 (e)	7	9.5.1, 9.5.6.3, 14.3b	H3, H6, H14

Question 33 — Biotechnology

Question	Marks	Content	Syllabus outcomes
33 (a) (i)	2	9.6.5.3.4, 9.6.5.2.2, 12.3c, 13.1e	H6, H12, H13
33 (a) (ii)	1	9.6.5.3.4, 9.6.5.2.2	H6
33 (b)	4	9.6.4.2.1, 13.1a, 13.1e	H6, H13
33 (c) (i)	2	9.6.1.2.2, 11.1b	H11
33 (c) (ii)	3	9.6.2.1, 14.1c, 14.3c	H6, H14
33 (d)	6	9.6.5, 9.6.6, 9.6.7	H4, H8
33 (e)	7	9.6.5, 9.6.6, 9.6.4, 14.3b	H3, H4, H8, H6, H14

Question 34 — Genetics: The Code Broken?

Question	Marks	Content	Syllabus outcomes
34 (a) (i)	2	9.7.6.2.1, 14.1a, 12.3c	H6, H9, H12, H14
34 (a) (ii)	1	9.7.1.2.1, 9.7.6.2.1, 12.3c, 14.1e	H12, H14
34 (b)	4	9.7.2.2.1, 9.7.2.2.2, 9.7.2.3.1, 13.1b, 13.1e	H9, H13
34 (c) (i)	2	9.7.7.2.1, 9.7.7.2.2, 9.7.7.3.1	H6
34 (c) (ii)	3	9.7.2.2.1, 9.7.7.3.1	H6, H4
34 (d)	6	9.7.2, 9.7.4	H4, H9
34 (e)	7	9.7.3, 9.7.4.2.3, 9.7.4.2.4, 14.3b	H2, H3, H6, H14

Question 35 — The Human Story

Question	Marks	Content	Syllabus outcomes
35 (a) (i)	2	9.8.2.2.4, 12.3c	H10, H13.1, H12.3
35 (a) (ii)	1	9.8.2.2.4	H10
35 (b)	4	9.8.3.2.1, 9.8.3.3.1, 9.8.2.2.5, 13.1b, 13.1e	H10, H13
35 (c) (i)	2	9.8.3.2.2, 9.8.5.2.1	H10
35 (c) (ii)	3	9.8.3.2.1	H10
35 (d) (i)	4	9.8.3.2.3, 9.8.4.1	H10
35 (d) (ii)	2	9.8.3.2.3, 9.8.3.3.3, 9.8.4.2.1, 9.8.4.3.1	H10
35 (e)	7	9.8.2.2.6, 9.8.2.3.1, 9.8.2.3.2, 14.3b	H3, H10

Question 36 — Biochemistry

Question	Marks	Content	Syllabus outcomes
36 (a) (i)	1	9.9.6.3.1, 13.1e, 12.3c	H6, H12, H13
36 (a) (ii)	2	9.9.6.2.3, 13.1e, 12.3c	H6, H13, H12
36 (b)	4	13.1b, 9.9.4	H6, H13
36 (c)	5	9.9.5, 11.3c	H2, H6, H11
36 (d)	6	9.9.2.2	H1, H6
36 (e)	7	9.9.3.2.2, 9.9.3.2.3, 9.9.3.3.4, 9.9.3.3.3, 14.3b	H3, H6, H14